

SOWJANYA PRIYA BANDARU

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SUMMARY

- Data Analyst with over 3 years of experience in data analysis, predictive modelling, and data visualization.
- Skilled in Python, SQL for data processing, cleaning, and developing machine learning models to support business decisions.
- Expertise in statistical analysis, A/B testing, time series analysis, and machine learning techniques such as Logistic Regression, Random Forest, and Naive Bayes.
- Experienced in creating interactive dashboards with Power BI and Tableau, delivering real-time business insights and visual analytics.
- Strong background in healthcare analytics, e-commerce, and customer service, optimizing processes and improving customer engagement.
- Proficient in AWS cloud platforms (Redshift, S3) and data engineering tools like Apache Airflow for seamless data integration and workflow automation.
- Well-versed in Natural Language Processing (NLP) for sentiment analysis and improving chatbot performance based on customer feedback.
- Demonstrated ability to build predictive models for fraud detection and patient care optimization, driving risk reduction and better outcomes.
- Collaborative and effective team player, working with engineers, healthcare providers, and marketing teams to align insights with strategic goals.

SKILLS

- **Programming Languages:** Python, SQL
- **Databases & Data Warehouses:** PostgreSQL, SQL Server, MySQL, Snowflake, Teradata, BigQuery
- **Data Visualization:** Power BI, Tableau, Advanced Excel
- **Libraries & Frameworks:** NumPy, Pandas, Matplotlib, Plotly, Seaborn, Streamlit, FastAPI, NLTK
- **Data Engineering:** Alteryx, Talend, Informatica, Apache Airflow, dbt
- **Cloud Platforms:** AWS (Redshift, DynamoDB, S3, Athena, Lambda)
- **Methodologies:** SDLC, Agile (Scrum), Waterfall
- **Version Control:** Git, GitHub
- **Machine Learning Techniques:** Regression, KNN, Decision Tree, Naive Bayes, Random Forest, LLM
- **Healthcare Technologies:** HIPAA, EMR, EHR
- **Analytical Skills:** Data Mining, Data Cleansing, Statistical Analysis, Data Wrangling, Time Series Analysis, A/B Testing, Text Mining, ETL, Data Warehousing, SAP, SAS

WORK EXPERIENCE

CVS Health

February 2025 – Present

Data Analyst

- Developed machine learning models to detect fraudulent activities by analysing historical claims data and transactional patterns.
- Pre-processed and cleaned large datasets using Python and SQL, ensuring high-quality data for training AI models and accurate fraud detection.
- Worked with cross-functional teams to integrate fraud detection models into the claims processing system, reducing false positives and improving detection efficiency.

CVS Health

July 2024 – December 2024

Data Analyst – Intern

- Created predictive models to identify at-risk patients and optimize care strategies, leveraging historical health data and patient demographics.
- Cleaned, pre-processed, and analysed large datasets from various healthcare systems with Python and SQL, ensuring high-quality data for model training and analysis.
- Utilized machine learning techniques such as Logistic Regression and Random Forest to build models that predict patient outcomes, enabling proactive intervention.
- Collaborated with healthcare providers and stakeholders to integrate predictive insights into patient care workflows, improving care quality and patient outcomes.

Dell Technologies

January 2021 – December 2022

Data Analyst

- Led the analysis of customer service interactions and designed data pipelines to enhance chatbot performance and ensure more accurate responses.

- Leveraged Python and SQL to clean and pre-process large datasets from customer service logs, enabling the chatbot to respond to user queries with higher efficiency.
- Created and implemented NLP (Natural Language Processing) models to analyse customer queries and improve the chatbot's understanding of context, resulting in a more intuitive user experience.
- Incorporated machine learning algorithms to continually optimize the chatbot's performance and accuracy by learning from real-time user interactions.
- Worked with cross-functional teams, including engineers and customer support, to ensure seamless integration of the chatbot into the existing customer service system.
- Evaluated customer feedback data to identify common issues and improve the chatbot's responses, enhancing overall customer satisfaction by 20%.
- Designed and maintained Power BI dashboards to track chatbot interactions, providing insights on user engagement, issue resolution time, and chatbot effectiveness.
- Conducted A/B testing to compare different versions of the chatbot, using statistical analysis to identify the most effective changes for better user interactions.
- Collaborated closely with business stakeholders to ensure the chatbot aligned with company goals, improving operational efficiency and reducing the workload on customer support agents.

TatvaSoft

March 2020 – December 2020

Data Analyst

- Analysed customer purchase patterns, browsing behaviour, and demographic data to create actionable insights for personalized marketing strategies.
- Leveraged Python, SQL, and EHR data to collect, clean, and transform large datasets from multiple e-commerce platforms, ensuring high-quality data for analysis.
- Designed and maintained interactive Tableau dashboards, providing real-time insights into customer trends and engagement, influencing key marketing decisions.
- Conducted detailed segmentation analysis using clustering techniques and customer cohort analysis to better target campaigns and improve customer retention.
- Implemented statistical techniques such as A/B testing to evaluate the effectiveness of various marketing strategies, resulting in a 15% improvement in conversion rates.
- Collaborated with cross-functional teams to integrate findings into the company's marketing and customer engagement workflows, enhancing personalization efforts.

THESIS

Efficient Federated Learning Algorithms for Resource - Constrained IoT Devices

Jan 2024 - Dec 2024)

- Conducted research on federated learning algorithms for IoT devices, focusing on energy efficiency, communication optimization, and privacy, while evaluating machine learning models.
- Developed scalable federated learning strategies for real-world applications like smart cities, analysing trade-offs between model complexity, accuracy, and resource utilization.

PROJECTS

Data Cleansing, Sales Analysis, and Dashboard Development (SQL, Power BI, Tableau)

- Developed a portfolio project showcasing SQL-based data cleansing, transformation, and sales analysis, ensuring data accuracy and consistency through advanced SQL functions
- Integrated an Excel sales budget dataset into Power BI for a dynamic Sales Dashboard and designed Tableau dashboards to analyse sales performance, customer trends, and key metrics.

CERTIFICATIONS

- Python for Data Science, AI & Development (Issued by Coursera)
- The Complete SQL Bootcamp: Go from Zero to Hero (Issued by Udemy)
- Tableau Ultimate course: From Zero to Hero (Issued by Udemy)
- Programming Essentials in Python (Issued by OpenEDG Python Institute)
- Python Basics (Issued by Coursera)

EDUCATION

Master of Science in Computer Science, California State University Northridge, Los Angeles, CA

January 2023 to December 2024

GPA: 3.77 / 4.0

Bachelor of Technology in Electronics and Communication Engineering, GVP college of Engineering, India

June 2018 to June 2022

GPA: 8.37 / 10.0